

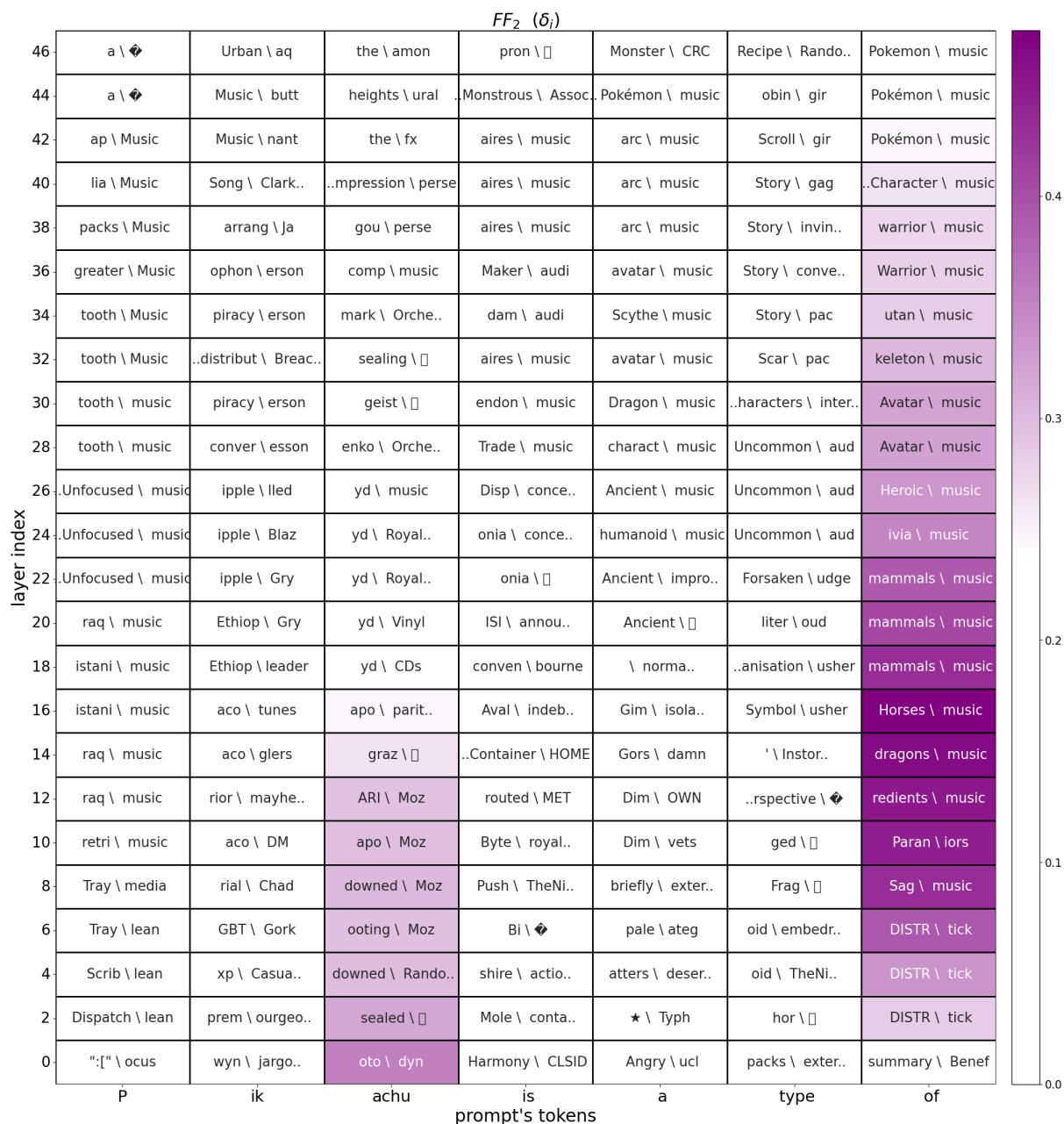


Figure 25: GPT2-xl FF_2 gradients via our spanning set interpretation and **normalized Logit Lens**. Every cell shows the most < probable \ improbable > tokens for its δ_i . Compared to Figure 13, we observe fewer non-English tokens in the projections and more tokens associated with the target token “music”.

I Editing Based on the “Shift” Mechanism

In Section 7 we introduce “forward pass shifting”, a method for LM knowledge editing through the approximation of the gradient matrix. In this section, we provide additional results and implementation details. We want to remind that our intention is not to propose a new alternative editing method, but rather to explore the concept of injecting knowledge into the model in an approximate manner to how backpropagation operates.

I.1 Comparison with Naive Backpropagation

We check our method’s ability to perform a single edit at a time of a given prompt, such that after editing, the most probable answer will be a selected target token. Our baselines for comparison are performing the same single Backpropagation editing with SGD and Adam. We use 50 samples from CounterFact (Meng et al., 2022) for the edited prompts and target. For each editing method and sample, we measure the method’s sensitivity to different learning rates by checking a range of possible values and finding the minimum one to achieve a successful edit. We also monitor the model’s general degradation after its update, using perplexity on maximum 256-tokens-long samples from WikiText (Merity et al., 2016). The findings are presented in Figure 26. Evidently, our approach’s minimum learning rate shows less variation compared to SGD. Furthermore, its stability to editing, as indicated by perplexity, surpasses that of Adam and SGD in earlier layers and is identical in the latter ones.

I.2 Benchmark Implementation Details

Based on the results from Section I.1, we identified layers 30–40 as the best potential editing layers. As for the learning rate, we examined values ranging from 0.14 to 0.26. The presented results use layer 35 with a learning rate of 0.24, which yielded the best results in the following benchmark.

The construction of the approximated gradient matrix is accomplished using the embedding of the target token. This embedding is obtained from the decoding matrix. If the target token consists of more than one token according to the model’s vocabulary, we select the prefix (the first token) to construct the target token.

Regarding the other editing methods and CounterFact benchmark implementation, we followed

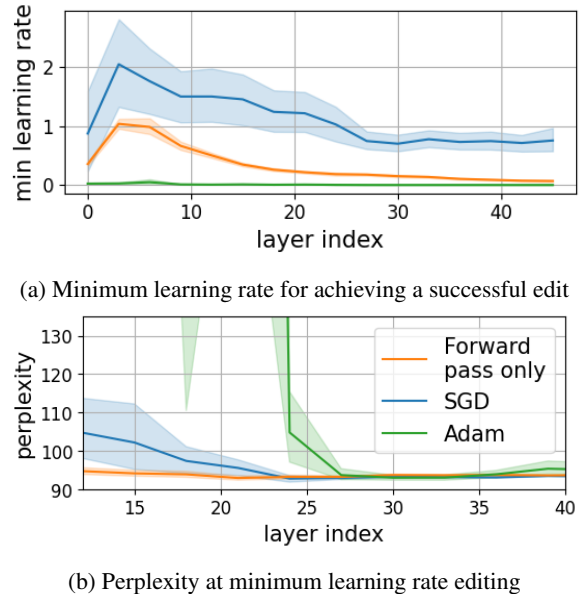


Figure 26: Editing only a single FF_2 matrix of GPT2-xl on CounterFact. By measuring a range of possible learning rate values we found the minimum that achieves a successful edit and measured the change in the model’s general knowledge through perplexity on WikiText.

the implementations provided by Meng et al. (2023) to create the results ourselves. The post-editing metrics included in this benchmark, which we present, are as follows: (1) “Efficacy” - the accuracy (percentage of times) with which the model predicts the targeted token as the most probable one given the editing prompt. (2) “Paraphrase” - the accuracy of the model to answer paraphrases of the edited prompt (also known as “Generalization”). (3) “Neighborhood” - the accuracy of the model on prompts from similar domains as the edited prompt, which we do not wish to change (also known as “Specificity”). (4) “N-gram entropy” measures the weighted average of bi- and tri-gram entropies, reflecting fluency level. In addition, we included in the benchmark the perplexity score of the 100 first sentences from WikiText (version wikitext-2-raw-v1, test split³) that are at least 30 characters long. Moreover, each prompt was truncated to its first 256 tokens. The score of this metric reflects how much the model’s original ability to generate text has changed.

The comparison was conducted on 1000 samples from CounterFact, and we benchmarked against state-of-the-art methods, such as ROME (Meng et al., 2022), MEMIT (Meng et al., 2023), and MEND (Mitchell et al., 2021).

³<https://huggingface.co/datasets/wikitext>

METHOD	EFFICACY $\uparrow$	PARAPHRASE $\uparrow$	NEIGHBORHOOD $\uparrow$
ORIGINAL MODEL	0.4 ± 6.31	0.4 ± 4.45	11.21 ± 19.77
FINE-TUNING (MLP 0)	96.37 ± 18.7	7.46 ± 19.44	10.12 ± 18.02
FINE-TUNING (MLP 35)	100.0 ± 0.0	46.1 ± 40.25	5.59 ± 13.19
MEND	71.4 ± 45.19	17.6 ± 31.47	7.73 ± 16.27
ROME	99.4 ± 7.72	71.9 ± 37.22	10.91 ± 19.53
MEMIT	79.4 ± 40.44	40.7 ± 41.64	10.98 ± 19.44
FORWARD PASS SHIFT	99.4 ± 7.72	41.55 ± 41.67	6.02 ± 14.0

Table 6: Single editing results of GPT2-xl on CounterFact benchmark

METHOD	N-GRAM ENTROPY $\uparrow$	WIKITEXT PERPLEXITY $\downarrow$
ORIGINAL MODEL	626.94 ± 11.56	93.56 ± 0.0
FINE-TUNING (MLP 0)	618.81 ± 52.74	199.84 ± 1392.66
FINE-TUNING (MLP 35)	618.5 ± 28.57	103.42 ± 11.69
MEND	623.94 ± 23.14	94.96 ± 0.66
ROME	622.78 ± 20.54	137.38 ± 786.35
MEMIT	627.18 ± 12.29	93.6 ± 0.11
FORWARD PASS SHIFT	622.45 ± 21.46	93.66 ± 1.08

Table 7: Examining the general ability of GPT2-xl to generate text after it has been applied with a single edit from CounterFact.

I.3 Results

In Table 6, we present the editing results regarding the model’s ability to modify internal knowledge. In Table 7, we analyze the effect of editing on the model’s ability to generate generic text.

If we filter out editing methods that cause drastic degradation to the model’s ability to generate text (where n-gram entropy falls below 620), our method achieves the best editing results on the edited prompt (Accuracy), tying only with ROME. With regards to editing the paraphrased prompts, the “forward pass shifting” method achieves comparable results, but falls behind ROME. However, one of our method’s limitations lies in its performance with neighborhood prompts, where the edited model altered prompts we do not anticipate to be changed.

I.4 Discussion

“Forward pass shift” demonstrates successful knowledge editing compared to methods that employ much more complex implementations. Additionally, our method shows minimal impact on the model’s ability to generate text, addressing one of the main challenges of fine-tuning in precise

knowledge editing.

From our understanding, “forward pass shift” is the simplest editing method in terms of algorithm complexity, as it only requires a single forward pass. The low score of this method in terms of neighborhood editing may be attributed to mistakenly editing activation patterns of FF_2 that are also shared with similar prompts. Future studies could explore the capability of “forward pass shift” to handle multiple edits at once, or to incorporate multiple iterations in its implementation (multiple forward passes).